

SHIPCS LOGISTICS

Negotiation Rules v1.0

Colaberry strawman for the Phase B deterministic negotiation engine. Mike Said reviews + redlines. Karun translates accepted rules into pure-function code per CLAUDE.md.

Built 2026-06-03 · BC ticket 9946715820 · Sources: May 21/26/28/29 call digests, Karun May 14 audit, v10 deck, dat-rfq-payload-schema.json

[FROM_SHIPCES] = stated on a call or in client docs

[INDUSTRY_DEFAULT] = standard freight-broker practice

[COLABERRY_PROPOSAL] = Colaberry default where no client guidance exists

1. Margin policy

Define the minimum, target, and aspirational margins the engine respects on every quote and every counter-offer.

Rule	Value	Tag
margin.absolute_floor_pct	7%	[INDUSTRY_DEFAULT]
margin.absolute_floor_usd	\$50 per load	[INDUSTRY_DEFAULT]
margin.target_pct	12%	[INDUSTRY_DEFAULT]
margin.aspirational_pct	18%	[INDUSTRY_DEFAULT]
margin.expedite_premium	+3 percentage points on expedite modes (target becomes 15%)	[COLABERRY_PROPOSAL] May 28 call: expedite is volatile, requires higher buffer
margin.cross_border_premium	+2 pp on US segment, 0 on MX segment	[COLABERRY_PROPOSAL] May 28: Mike said Mexican rates do not fluctuate
margin.below_floor_action	WALK (log walk-away with reason=below_floor)	[INDUSTRY_DEFAULT]
margin.below_target_above_floor	PROCEED with audit flag; daily margin report shows these	[COLABERRY_PROPOSAL]

Pure-function gate (pseudocode):

```
function gateMargin(sell_rate, buy_rate, service_mode, is_cross_border) -> Gate:
  target_pct = 12
  if service_mode in {EXPEDITE_SOLO, EXPEDITE_TEAM, EXPEDITE_EXCLUSIVE}:
    target_pct += 3
  if is_cross_border:
    target_pct += 2 # only on US-side legs

  margin_pct = (sell_rate - buy_rate) / sell_rate * 100
  margin_usd = sell_rate - buy_rate

  if margin_pct < 7 or margin_usd < 50:
    return Gate(action="WALK", reason="below_floor")
  if margin_pct < target_pct:
    return Gate(action="PROCEED", audit_flag="below_target")
  return Gate(action="PROCEED")
```

2. Quote pricing inputs (rate basis)

The engine uses these sources in priority order. Selection rule: first source returning a non-null value wins; log as `rate_basis.source` on the quote.

Priority	Source	When applied	Tag
1	Customer rate table match on lane + service type	Customer-specific contract rate exists for the lane	[FROM_SHIPCES] Karun May 14 audit GTM section 4
2	Historical Colaberry closed-deal average last 30 days	At least 3 closed deals match the lane	[COLABERRY_PROPOSAL] builds on BC 9946715790 internal loop
3	DAT Rate View 7-day average for lane + equipment	DAT API credentials + 7-day data exist	[FROM_SHIPCES] May 28: Mike standard practice
4	DAT Rate View 30-day average	Fallback when 7-day is empty	[FROM_SHIPCES] May 28 call
5	DAT manual scrape spot rate	DAT API unavailable, scrape session live	[COLABERRY_PROPOSAL] BC 9946715770 due 2026-06-07
6	HITL escalation	None of the above resolve	[INDUSTRY_DEFAULT]

3. Service-mode pricing modifiers

Translate base lane rate to service-mode-specific quote. Per Karun D5: when an RFQ qualifies for multiple modes, emit one quote option per mode (Brett Pair A/B/C, slides 11-14).

Mode	Modifier on base lane rate	Tag
FTL (Truckload)	x 1.00	[INDUSTRY_DEFAULT]
LTL	x 0.60 + LTL classification fee	[INDUSTRY_DEFAULT]
ELTL	x 0.85	[INDUSTRY_DEFAULT]
PARTIAL	x 0.70	[INDUSTRY_DEFAULT]
EXPEDITE_SOLO	x 1.35	[INDUSTRY_DEFAULT] expedite spot premium
EXPEDITE_TEAM	x 1.75	[INDUSTRY_DEFAULT] team drives 2x linehaul speed
EXPEDITE_EXCLUSIVE	x 2.10 (single-load truck, no co-load)	[INDUSTRY_DEFAULT]
INTERMODAL	x 0.55 + ramp fees	[INDUSTRY_DEFAULT]

4. Special-requirements premiums

Requirement	Premium	Tag
Hazmat (class 3-9)	+\$300 flat , capped at +8% of base	[INDUSTRY_DEFAULT]
Hazmat class 1 (explosives) / class 7 (radioactive)	+\$800 flat , no cap, manual approval required	[INDUSTRY_DEFAULT]
Team driver (separate from EXPEDITE_TEAM mode)	+\$0.40 per mile	[INDUSTRY_DEFAULT]
Liftgate	+\$75 flat per stop	[INDUSTRY_DEFAULT]
Inside delivery	+\$150 flat	[INDUSTRY_DEFAULT]
TWIC	+\$100 flat	[INDUSTRY_DEFAULT]
TSA	+\$150 flat	[INDUSTRY_DEFAULT]
FAST card (cross-border)	+\$0 (table stakes for MX cross-border)	[COLABERRY_PROPOSAL]
Temp control (reefer)	+12% of base	[INDUSTRY_DEFAULT]
High-value cargo (>\$100K declared)	+\$200 flat + insurance verification	[INDUSTRY_DEFAULT]

Requirement	Premium	Tag
Tarp	+\$50 flat	[INDUSTRY_DEFAULT]
Driver assist (load/unload)	+\$100 flat per stop	[INDUSTRY_DEFAULT]

5. Cross-border specific rules (US-MX)

Lock the two-segment pricing pattern Mike described on May 28.

Rule	Value	Tag
cross_border.us_segment_modifier	Use US rate basis (Section 2), apply Section 3 mode modifier	[FROM_SHIPCES] May 28 call
cross_border.mx_segment_modifier	Use MX rate table (BC 9946715804 due 2026-07-12); no volatility premium	[FROM_SHIPCES] May 28 verbatim: "Mexican rates don't fluctuate"
cross_border.border_crossing_buffer_hours	18 hours per geofence zone (matches BC 9951047176 geofencing approach)	[COLABERRY_PROPOSAL] Brett May 28 demo: Cadereyta to Laredo = 18 hours
cross_border.customer_default_crossing	Use customer override if present, else nearest geographic crossing	[INDUSTRY_DEFAULT]
cross_border.requires_fast_card_carriers	TRUE for southbound and northbound	[COLABERRY_PROPOSAL] MX cross-border best practice

6. Initial offer to carrier (sourcing side)

Rule	Value	Tag
carrier_initial_offer_pct_of_quoted	88% of customer sell rate (= 12% margin baked in)	[COLABERRY_PROPOSAL] aligns with margin.target_pct
carrier_initial_offer_floor	= sell rate * (100 - 7) / 100	[INDUSTRY_DEFAULT]
carrier_initial_offer_disclosed	FALSE on DAT postings; show only on direct outreach	[INDUSTRY_DEFAULT] May 28: most DAT postings hide the rate
carrier_tier_bonus_t1_over_floor_usd	+\$200 for T1 (pre-approved, lane-trusted)	[COLABERRY_PROPOSAL]
carrier_tier_bonus_t2_over_floor_usd	+\$100 for T2 (pre-approved, no lane history)	[COLABERRY_PROPOSAL]
carrier_tier_bonus_t3_over_floor_usd	+\$0 for T3 (random DAT, no history)	[COLABERRY_PROPOSAL]

7. Counter-offer policy (carrier side)

Rule	Value	Tag
negotiation.max_rounds	3 rounds	[INDUSTRY_DEFAULT]
negotiation.round_1_concession_pct	+3% of initial offer if carrier counter within 15% of our offer	[COLABERRY_PROPOSAL]
negotiation.round_2_concession_pct	+2% if still negotiating	[COLABERRY_PROPOSAL]
negotiation.round_3_concession_pct	+1% (final); walk if not accepted	[COLABERRY_PROPOSAL]
negotiation.walk_trigger_margin_floor	Walk if next offer would breach margin.absolute_floor_pct	[INDUSTRY_DEFAULT]
negotiation.walk_trigger_time_minutes	45 minutes inactivity on expedite, 6 hours on standard	[COLABERRY_PROPOSAL]
negotiation.walk_trigger_alterate_found	Walk immediately if another carrier accepts our open offer on same load	[INDUSTRY_DEFAULT]
negotiation.escalate_to_human_at_round	Escalate to dispatcher at round 3 if margin within 1 pp of floor	[COLABERRY_PROPOSAL]

Pure-function pseudocode:

```

function nextOffer(round, prior_offer, carrier_counter, sell_rate) -> Decision:
  if round > 3:
    return Decision(action="WALK", reason="max_rounds")

  concession = {1: 0.03, 2: 0.02, 3: 0.01}[round]
  candidate_offer = prior_offer * (1 + concession)

  if carrier_counter > sell_rate * 0.93: # carrier wants more than 7% off sell
    return Decision(action="WALK", reason="below_floor")

  if candidate_offer < carrier_counter:
    candidate_offer = min(carrier_counter, sell_rate * 0.93)

  margin_pct = (sell_rate - candidate_offer) / sell_rate * 100
  if margin_pct < 8 and round == 3:
    return Decision(action="ESCALATE_HUMAN", reason="thin_margin_final_round")

  return Decision(action="COUNTER", new_offer=candidate_offer, round=round)

```

8. Customer-side rules (quote validity + re-quote)

Rule	Value	Tag
quote.validity_expedite_hours	2 hours	[INDUSTRY_DEFAULT]
quote.validity_standard_hours	24 hours	[FROM_SHIPCES] May 28: 1-2 week validity on RFQ batches
quote.validity_rfq_batch_days	7 days with 48-hour notice required	[FROM_SHIPCES] May 28 verbatim: "valid for 1 week, 2 weeks, 48 hours notice"
quote.requote_market_move_threshold_pct	Re-quote if DAT 7-day avg moves more than 10%	[COLABERRY_PROPOSAL]
quote.requote_time_threshold_pct_of_validity	Re-quote at 80% of original validity if not awarded	[COLABERRY_PROPOSAL]
quote.discount_for_re_engagement_max_pct	Max 3% discount off original quote when re-engaging same customer same lane within 24h	[COLABERRY_PROPOSAL]
quote.multi_option_emit	One quote option per qualifying service type per Karun D5	[FROM_SHIPCES] Karun D5 May 13

9. Customer-specific overrides (CustomerRule profile)

Maps to customerRules block on the canonical RFQ payload (BC 9917948332).

Customer attribute	Engine behavior	Tag
isBoardEmail = TRUE	Downgrade priority by 1 tier; do not auto-quote	[FROM_SHIPCES] Karun May 14 audit
kpiDriven = TRUE	Tag for weekly \$ report; prioritize sourcing	[FROM_SHIPCES] Ram KPI directive May 21
premiumPricing = TRUE	Margin floor +2 pp; target +4 pp	[COLABERRY_PROPOSAL]
competitionCount > 3	Tighten margin: target = floor + 2 pp	[COLABERRY_PROPOSAL]
appointmentsRequired = TRUE	Add liftgate + dock check on every stop validator	[INDUSTRY_DEFAULT]
fuelSurchargeConfig.model = INDEX_LINKED	Fuel rides separately; do not bake into all-in	[INDUSTRY_DEFAULT]

10. Escalation matrix (when the engine refuses to act autonomously)

Trigger	Route to	Tag
Margin floor breached on initial pricing	Dispatcher	[INDUSTRY_DEFAULT]
Margin floor breached on round 3 counter	Dispatcher	[INDUSTRY_DEFAULT]
Customer requests discount > 5% off quoted	Account owner	[COLABERRY_PROPOSAL]
Carrier counter > sell rate x 0.93 (below 7% margin)	Dispatcher	[INDUSTRY_DEFAULT]
Market move > 10% post-quote, pre-award	Account owner	[COLABERRY_PROPOSAL]
Hazmat class 1 or 7 detected	Compliance + Dispatcher	[INDUSTRY_DEFAULT]
Cross-border without FAST-card carrier	Cross-border specialist (Monterey HITL per BC 9951047130)	[FROM_SHIPCES] May 28 call
3 consecutive walk-aways on same load	Dispatcher + Account owner	[COLABERRY_PROPOSAL]
Confidence score on RFQ extraction below 0.65	HITL queue with reason "extraction"	[FROM_SHIPCES] Karun V3 validator

11. Audit and observability requirements

Each negotiation event emits one structured log row. Daily margin report (broker-owner-facing) shows margin distribution by lane, mode, customer class, carrier tier; surfaces below-target deals so the rules can be tuned.

```
{
  "event_id": "evt_01JV9R00000000000000000000XX",
  "timestamp": "ISO-8601",
  "rfq_id": "rfq...",
  "load_id": "ld...",
  "carrier_id": "car...",
  "event_type": "INITIAL_OFFER | COUNTER | ACCEPT | WALK | ESCALATE",
  "round": 1,
  "our_offer": 2200.00,
  "carrier_counter": 2250.00,
  "sell_rate": 2500.00,
  "margin_pct_at_close": 12.0,
  "rate_basis_source": "dat_rate_view_7day | customer_rate_table | historical_30d | manual_scrape | hitl",
  "decision_code": "COUNTER_ROUND_1 | WALK_BELOW_FLOOR | ESCALATE_THIN_MARGIN | ACCEPT_AT_TARGET",
  "rule_versions": { "negotiation_rules": "v1.0", "margin_policy": "v1.0" }
}
```

12. Out of scope for v1.0

- Spot-rate market arbitrage (buy-load-sell-load triangulation). Revisit Q4 2026 after internal historical loop has data.
- Multi-load contract bidding (annual RFQs with hundreds of lanes). Separate Phase D effort.
- Real-time fuel surcharge index integration (DOE weekly). Covered by Pricing-rules engine BC 9946715801 due 2026-07-05.
- Carrier scoring weights (which carrier to call first). Separate spec, BC 9946715826 due 2026-06-06.
- Counter-offer language generation (the actual email reply). Belongs to the response generator agent, not the rules engine.

13. Open questions for Mike's review

Specific items where ShipCES input replaces a [COLABERRY_PROPOSAL] or [INDUSTRY_DEFAULT] tag. Reply inline on each numbered question lands as v1.1; a complete written rules doc lands as v2 and supersedes this.

1. What is ShipCES current absolute margin floor in percent? (Section 1)
2. What is the target margin per service tier? (Section 1)
3. Margin uplift for cross-border vs domestic - confirm 2 pp US-side, 0 MX-side? (Section 1, 5)
4. Default initial-offer percentage of sell rate - 88% reasonable? (Section 6)
5. Walk-away inactivity timer on expedite - is 45 minutes right or longer? (Section 7)
6. Quote validity for standard service - 24 hours plus 1-2 week RFQ-batch validity, both apply? (Section 8)
7. Hazmat class 1 and 7 auto-escalate - confirm manual-only? (Section 4, 10)
8. Discount cap on customer re-engagement - 3% off original quote acceptable? (Section 8)
9. Are there customer-specific rules already in your current process that we should encode in the CustomerRule profile? (Section 9)
10. Are there carrier-tier definitions (T1/T2/T3) you already use, or do we propose ours? (Section 6)

Built by Colaberry for ShipCES Autonomous Brokerage · 2026-06-03 · BC ticket 9946715820 · em-dash count 0 · ShipCES rules supersede on receipt as v2.